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Schoolyard Level Inventory for Describing the Environment: Linking High-Resolution Spatiotemporal Data to the Physical Environment to Understand Children's Physical Activity Behaviors

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ABSTRACT

Background: Schoolyards and/or playgrounds are important settings for fostering children's social, cognitive, and physical development. Optimal schoolyard design is likely an important health-promoting and equigenic strategy, but prior research does not fully consider multiple dimensions of these spaces, such as vegetation, slopes, ground cover, and equipment.

Aims: We describe the Schoolyard Level Inventory for Describing the Environment (SLIDE) a novel method to compile spatial data from multiple sources, including in-person audits (e.g., cleanliness), images (e.g., greenness), and geographic information systems (e.g., distance to road), that can be linked with children's device data.

Materials & Methods: We illustrate use of SLIDE in 91 first and second graders from three elementary schools who wore an accelerometer and global positioning system (GPS) device during school recess. Multilevel linear regression was used to test associations between 22 environmental features and children's physical activity intensity and time spent (duration of play).

Results: The auditing process took two researchers ~30 min per 1000 m² of schoolyard. Some features were consistent across schools (flatness, grass areas, views of the school), while other features varied (views of the road, covered areas, seated areas, slides, and colorful equipment). We found significant, independent, negative associations between physical activity intensity with redness and cleanliness, and positive associations with colorful equipment and distance to road. We detected significant, negative associations between time spent with greenness and fields, and a positive association with equipment condition.

Discussion: While many of our findings align with prior research, we note that SLIDE also provides new insights including a better understanding of the complex inter-relationships between schoolyard features and outcomes. On a larger sample of more diverse schoolyards, this would allow researchers to identify which specific features should be the focus of future schoolyard designs or renovations.

Conclusion: SLIDE can be used to inform schoolyard design, assess greening or redesign interventions, and examine schoolyard inequalities, while accounting for complex inter-relationships between features.

1 | Introduction

Schoolyards and/or playgrounds are important settings for fostering children's social, cognitive, and physical development [1–4]. In addition to being the most common physical activity amenity in children's neighborhoods [5, 6], the schoolyard is the primary location for school recess, children's only opportunity for unstructured free-play during the otherwise structured and sedentary school day. During recess, children may be physically active, practice motor skills, socialize, interact with nature, be autonomous by selecting how and with whom they spend their time, assess risk and consequence, and learn to create and follow rules [7–9]. In fact, physical activity during recess constitutes up to 70% of daily physical activity [10]. However, schoolyards vary in size, content, and condition, and these design aspects influence children's opportunity for positive behaviors like physical activity and the likelihood of antisocial behaviors like bullying [11–16]. There are lifelong effects of childhood health [17], physical activity [18], and positive and negative recess experiences [19], so optimizing the schoolyard experience for all is imperative for both short- and long-term well-being and reductions in child health disparities.

Optimal schoolyard design is a potentially important health-promotion and equigenic strategy due to the number of children who could be influenced by an intervention of this type, with little ongoing time or monetary burden to school staff or the community [12, 13, 20–24]. Full schoolyard redesigns are costly, so evidence-based design is crucial to optimizing the use of limited funds. However, other schoolyard design features may be modifiable (e.g., tree planting) or mobile (re-location of fixed equipment) at much lower costs. While we will focus here on the association of schoolyard design with physical activity and time spent, schoolyard redesign can positively influence children in other ways, for example, by reducing children's exposure to heat, noise, or air pollution [25–30], discouraging antisocial behaviors [14], and improving mental health and restoration [31]. Further, schoolyard redesign may be used to reduce inequities in physical activity [12], access to natural or green environments [32, 33], and environmental exposures [34] between groups. For example, children in poorer neighborhoods and those with more Black residents have lower quality and less safe playgrounds [35, 36]. There are also differences in where boys and girls are physically active, with some schoolyard settings (fixed equipment, sport-specific areas) ameliorating the commonly seen sex difference in physical activity participation [12, 37, 38].

Researchers aiming to identify salient features of the schoolyard environment have used direct observation [21, 37, 39–43] or a combination of Global Positioning System (GPS) devices and accelerometers or heart rate monitors to identify where children are spending their time or being physically active [38, 44–48]. This research tends to evaluate one feature at a time and has generally found that children are most active when located on fields [49, 50], fixed equipment [49, 51, 52], and blacktop or asphalt courts [39] although there is variability across schoolyards and by demographic characteristics (e.g., age, sex) [37, 39, 42, 47, 48, 52] that necessitate more research with larger samples on many diverse schoolyards. The primary limitation of this prior research is that it fails to consider the complex relationships *among* schoolyard

features. For example, only considering the presence/absence of fixed equipment ignores potentially useful information regarding aspects such as ground cover or slope, availability of shade, proximity to other features, or the size, colorfulness, or singular or multi-component nature of equipment.

A related issue is that existing approaches to characterizing the schoolyard environment similarly do not consider the co-occurrence of potentially relevant information. For example, audit tools often include checklists for the presence/absence of specific features (e.g., slides, trees) but no corresponding information further characterizing these features, such as their condition or colorfulness [53–57]. Further, existing audit tools ignore potential heterogeneity in features across the schoolyard by, for example, only including a single 'cleanliness' value for the entire playground [54]. Lack of information about the spatial location of features also precludes linkage with other data sources like device-based measures of children's location and activity intensity. Thus, a spatially-linked audit method would provide more detailed information about the schoolyard environment, where children spend time and/or participate in physical activity, and the complex relationships amongst the schoolyard features. Such a method could also usefully quantify changes to the schoolyard pre- and post-intervention and quantify important inequalities in schoolyard quality across a city or region, which can be used to target resources and intervention efforts. We demonstrate the potential utility of SLIDE by demonstrating how it can be used to study associations between audited features and children's physical activity intensity and time spent using a sample of first and second graders across three elementary schoolyards.

2 | Methods

2.1 | Participants and Setting

The Michigan State University institutional review board approved this protocol (STUDY00002552). Three elementary schools near Lansing, Michigan agreed to participate in this study, and information was provided to parents of all children in first or second grade. These grades were selected because the original intention was to monitor changes in the children's physical activity and schoolyard renovations over time, a process that would have taken 3 years (so the second graders would then be in fifth grade whereas older children would have left the school by that time). However, all schools closed the following year due to the COVID-19 pandemic, so no additional data collection occurred after the baseline data collection (used in the present analysis). Consent from one parent/guardian and written assent from each child was provided prior to data collection, which occurred in May/June 2019 from 10 am to 2 pm.

2.2 | Audit Development and Implementation

Audits can be conducted both in-person and remotely using aerial imagery or Google Street View imagery [58] and have been developed to reliably quantify numerous aspects of the physical environment, including walkability [59], disorder [60], environmental aesthetics [61], and smokefree signage at schools [62]. The actual content of the audit will depend on the purpose and/or

research question. The SLIDE audit is a system of collecting and synthesizing multiple streams of spatially linked information, such as in-person audits, images, and geographic information systems (GIS). The specific measures described below are not required of all SLIDE users, and we are not advocating for their use specifically; rather, we use the collected variables as an example of how SLIDE works in practice. The research team developed this specific SLIDE audit based on established audit systems for rating the overall schoolyard environment [53–57] and prior research relating features of the schoolyard physical environment to outcomes [11]. SLIDE is meant to be flexible, so other users may include different variables or data streams than described below (further described in the discussion).

The in-person portion of the audits was conducted using the ArcGIS Collector application (ESRI, Redlands, CA) on an iPad (Apple, Cupertino, CA). ArcGIS Collector has since been replaced with ArcGIS Field Maps, which has the same functionality. Prior to data collection, we created a fishnet grid with 7 m cells for each schoolyard using ArcPro. Two trained coders physically stood in the centroid of each cell and coded five characteristics including the gradient (sloped or flat), ground cover (e.g., grass, mulch, fixed rubber), and whether or not the playground equipment, school building, or road were in view (Figure 1). All codes were agreed upon in real time by both coders. These codes were based on the Environmental Assessment of Public Recreation Spaces (EAPRS) tool, except for views of the equipment, school building, and road which were selected because of their theoretical influence on children. For example, a child might feel or actually be less supervised when they cannot see the school

directly so they can do different things like talk more openly with their friends, participate in risky play, or be in nature. Viewing the playground equipment may attract children to that location or promote activity as they see others play or options for play.

A 360° image was taken by the authors within each grid cell using a 15 megapixel Samsung Gear Camera (Samsung, Seoul, South Korea), which can be converted to two flat 180° images using GIMP 2.10 software (GIMP, Charlotte, NC). For each image, color saturation was quantified to derive proportion of pixels that were predominantly red, green, or blue using a bespoke R tool (available upon request). Specifically, for each pixel in each image, the predominant band (red, green, or blue) was assigned if the average saturation of the other two bands is smaller than the saturation of the predominant band. Average redness, greenness, and blueness across the two images was calculated for each grid cell. Laboratory evidence has shown an increase in parasympathetic nervous activity and lowered heart rate when viewing images of trees [63, 64], a preference for more ‘natural’ imagery [65], and a preference for green and blue hues, and ‘natural’ compared to built-up features [66–68]. This may be important as understanding how where children spend their recess time (and how green that area is) may be important in studies of the relationship between physical activity and behavioral or psychological issues.

In addition to the centroid data, we recorded ancillary data for specific schoolyard features, which could either be points or polygons. Specifically, we used ArcCollector to record the location of swings, slides, seating, sports fields, hard surface areas, and general points or polygons (i.e., “other”). For each feature, we

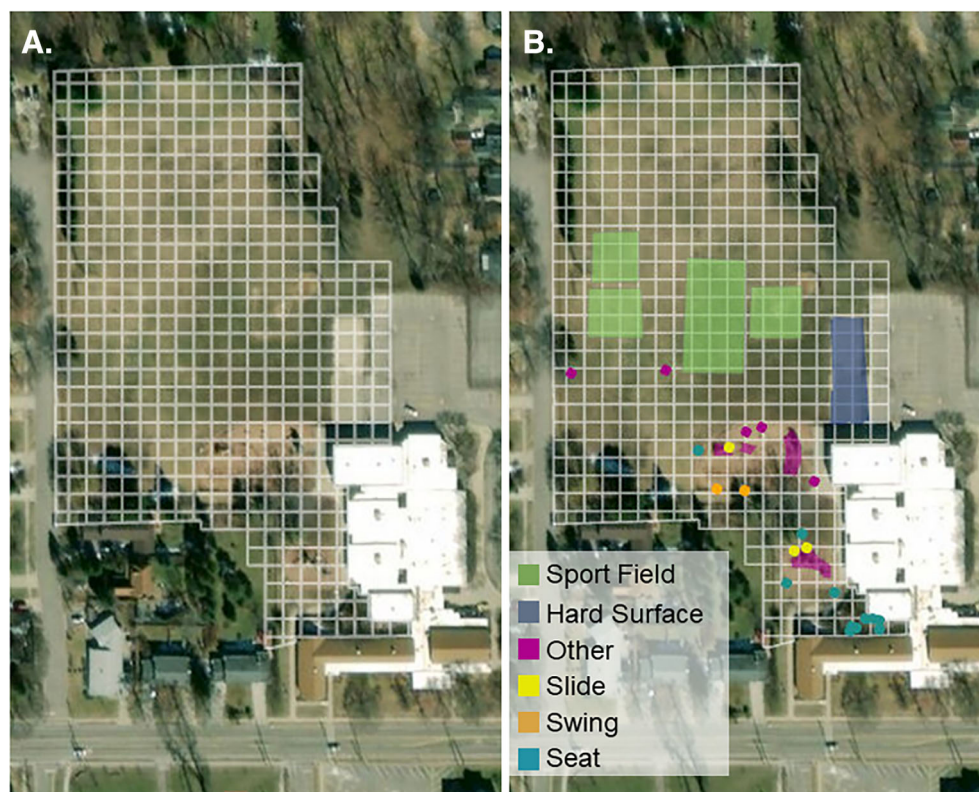


FIGURE 1 | Grid (A) in which gradient, ground cover, and whether or not the playground equipment, school building, or road were in view were recorded and feature points and polygons (B) which were later merged with the grid.

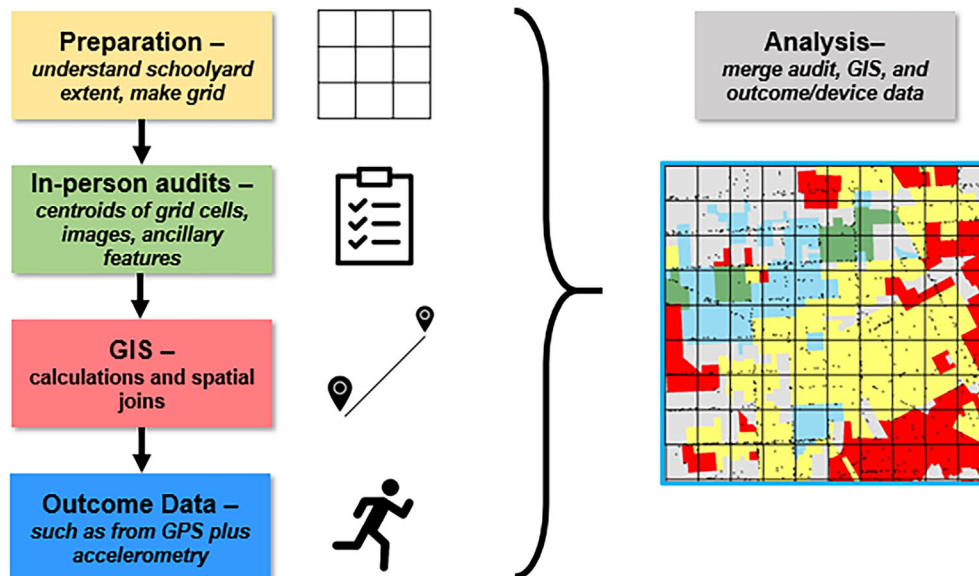


FIGURE 2 | Overview of data linkage process for SLIDE.

coded the condition on a scale from 1 (lack of function) to 3 (completely functional and in good working condition), cleanliness on a scale from 1 (very unclean or poorly maintained, peeling paint) to 3 (in good aesthetic condition, free of dirt/debris, painted, only showing signs of gentle use), the number of access points (i.e., entrances to a play structure), and whether the feature included a built-in cover and/or was colorful (2+ non-neutral colors). For hard surface and field areas, we recorded whether there were painted markings present (e.g., four-score board). We also included additional information that is not being used in the present analysis, such as the height, equipment material, seating or swing type, and whether the space was usable or not. These codes were primarily based on the EAPRS and our knowledge regarding common spaces found on schoolyards [12]. Cleanliness has previously been related to the use of specific schoolyard spaces [69]. A number of factors go into deciding the colorfulness of playground equipment (e.g., desired aesthetics). Children prefer colorful equipment [70] and color affects mood and behavior [71, 72], so could reasonably influence children's play.

Last, we calculated some geographic variables using GIS in ArcPro. Specifically, we separately calculated the distance (meters) from each grid centroid to both the school building and the nearest road. Distance to the school has previously been related to physical activity [73], while the road may influence noise, pollution, or safety. All data were then linked using spatial joins so that each grid cell contained information regarding the general descriptors, image data, additional features like swings or slides, and distance calculations for each grid centroid. These linked data (Figure 2 for overview of process) were used to calculate several exposure variables, which are described in Table 1.

2.3 | GPS and Accelerometry Data Collection on Children

Participating children ($n = 91$) in each school ($n = 3$) were fitted with an elastic belt to position a GPS and accelerometer over

their right hip at the start of all regularly scheduled recess periods for up to 4 days (one school was only available for 3 days of data collection). Recess periods were scheduled to be 20 min in duration and took place twice a day (morning and afternoon), weather permitting (1–2 recess periods per school were canceled due to rain). Overall, there was a median of 3 days (range 1–4), with a median of 5 recess periods (range 2–7) per child. Missing recess data were due to child absences unrelated to the study.

The GPS units were initialized to record the time and position (latitude, longitude) every second. The GPS was either a Canmore GT-730FL-S (TSKY Co., Zhubei City, Taiwan) or a QStarz BT-Q1300ST (QStarz International Co., Taipei, Taiwan), which was configured using the CanWay (version 1.1.12) or QSports (version 3.76) software, respectively. Accelerometers were ActiGraph GT3X+, GT3X-BT, or GT9X triaxial accelerometers (ActiGraph LLC, Pensacola, FL) initialized using ActiLife software (version 6.13.4) to collect raw acceleration data at a sampling rate of 30 Hz. Different ActiGraph models were used due to the limited number of devices available; these monitors have been previously shown to record similar acceleration data [74, 75].

Each day, the belts were collected, and data were downloaded. Accelerometer data were used to calculate vector magnitude counts $\cdot s^{-1}$, which is the square root of the sum of the squared counts in each axis. This is a commonly used metric of physical activity intensity, and the short epoch is used to capture the transient nature of children's movements [76, 77]. GPS and accelerometer data were linked based on exact matches between timestamps in RStudio (Vienna, Austria; version 1.3.1056). A small number (3%) of GPS points were removed from the analysis because they did not overlap with our audit area. The outcomes of interest were mean vector magnitude across all participating children and the number of points (i.e., time spent) within each grid cell.

TABLE 1 | Exposure variables calculated from audit data.

Exposure	Source
Gradient	Centroid
Ground cover	Centroid
Playground equipment view	Centroid
School building view	Centroid
Road view	Centroid
Mean greenness	360 image
Mean redness	360 image
Mean blueness	360 image
Condition	Mean of all features in a cell, NA if no features
Cleanliness	Mean of all features in a cell, NA if no features
Colorfulness	Yes/no if any feature in a cell was colorful (2 or more colors), NA if no features in the cell
Cover	Yes/no if any feature in a cell was covered, NA if no features in the cell
Access points	Total number of access points across all features in a cell, NA if no features in the cell
Markings	Yes/no presence of markings on hard surface or field features, NA if these features were not present in the cell
Swing	Yes/no presence of a swing feature
Seat	Yes/no presence of a seat feature
Slide	Yes/no presence of a slide feature
Hard surface	Yes/no presence of a hard surface area
Field	Yes/no presence of a field area
Other point or polygon	Yes/no presence of a general point or polygon
Distance to school	Calculated using centroid and location of school
Distance to road	Calculated using centroid and location of road

Abbreviation: NA, not applicable.

2.4 | Statistical Analyses

Descriptive statistics for audit features were calculated by school and overall. Then, the associations between each audit feature and mean vector magnitude and the number of points within each grid cell were assessed separately using a multilevel linear regression wherein grid cells were nested within schools. In addition to individual models in which each exposure was separately related to each outcome, we also tested models in which all exposures were entered into the model together to be related to each outcome. Both modeling approaches were used to align with prior research (which considers individual environmental predictors) and to gain a preliminary understanding of the inter-relationships between environmental features. Due to skew, the number of data points in each cell were log transformed in all models. Analyses were conducted using the lme4 package (version 1.1–26) in RStudio.

3 | Results

Children ($n = 91$, 69 girls) were 7–9 y of age (mean $\pm$ SD: 7.8 ± 0.7 y) and predominantly White (80%) (data not shown in tabular form). The audit process was conducted by two researchers (ALP and KAC) for the three schools over about 40 h. Sizes of the schoolyards were relatively large (mean $29,582 \text{ m}^2$). In other words, one might expect this process to require two researchers

30 min per 1000 m^2 of schoolyard. Average duration of recess was 26.9 ± 5.6 min. Post-processing of audit data consisted of linking feature and grid data, checking for consistency in recorded data, and calculating outcome variables (e.g., spatially-derived variables, greenness, mean cleanliness for all features in a cell) which took approximately 6 h for all three schools. Post-processing of GPS plus accelerometer data involved timestamp linkage, removing data points that fell outside the recess period or schoolyard, and joining with audit data, which took approximately 2 h.

Audit features are reported in Table 2 and illustrated for one school in Figures S1–S9. Across the schools, schoolyards were predominantly flat (95%), covered by grass (87%), and had clear views of the school ($> 90\%$). Variation across the schools was observed, namely in views of the road, amount of covered areas, seated areas, slides, and colorful equipment. Small differences in greenness, redness, or blueness of views and in distance to the school (from each cell) were detected across the schools. Larger differences were observed for cleanliness, condition, and access points of playground equipment and distance to road. Outcome variables (mean vector magnitude and time spent within grid cells) are reported in Table 2 and illustrated for one school in Figure 3.

Regression modeling separately detected significant negative associations between physical activity intensity (mean

TABLE 2 | Audit outcomes for each school and overall.

Exposures	Overall Percent	School 1 Percent	School 2 Percent	School 3 Percent
Gradient				
Sloped	5.1	5.2	4.6	5.6
Flat	94.9	94.8	95.4	94.4
Ground cover				
Asphalt	2.6	2.8	1.6	3.9
Brick	0.2	0.0	0.0	0.8
Concrete	4.2	3.2	4.5	5.2
Dirt	0.7	0.3	0.0	2.3
Grass	86.5	87.5	90.6	79.0
Mulch	4.5	2.7	3.3	8.9
Fixed rubber	0.1	0.2	0.0	0.0
Sand	1.1	3.3	0.0	0.0
Playground equipment view	96.5	100.0	91.5	99.6
School building view	96.6	100.0	91.5	100.0
Road view	73.5	100.0	40.6	89.9
Cover	30.2	16.0	58.3	31.8
Markings	11.0	100.0	100.0	29.4
Swing	1.0	0.5	0.5	2.5
Seat	2.9	1.6	0.5	8.5
Slide	1.4	0.3	0.4	4.3
Hard surface	7.0	5.9	6.5	9.3
Field	15.3	12.7	11.9	23.9
Other point or polygon	4.4	2.2	0.9	12.6
Colorful	29.4	40.0	100.0	27.1
	Mean ± SD	Mean ± SD	Mean ± SD	Mean ± SD
Greenness	0.5 ± 0.1	0.5 ± 0.1	0.6 ± 0.1	0.4 ± 0.1
Redness	0.1 ± 0.1	0.1 ± 0.1	0.2 ± 0.1	0.0 ± 0.0
Blueness	0.3 ± 0.1	0.4 ± 0.0	0.3 ± 0.1	0.3 ± 0.1
Condition (range 1–3)	1.7 ± 0.7	2.3 ± 0.6	1.3 ± 0.5	1.5 ± 0.5
Cleanliness (range 1–3)	2.0 ± 0.4	2.4 ± 0.5	2.0 ± 0.1	1.8 ± 0.3
Access points	2.1 ± 4.6	0.9 ± 2.6	0.9 ± 2.8	3.6 ± 5.9
Distance to school (m)	87.6 ± 52.0	86.1 ± 47.1	97.1 ± 57.7	74.9 ± 45.7
Distance to road (m)	77.8 ± 37.3	74.6 ± 38.5	96.8 ± 30.6	52.4 ± 27.5
Outcomes	Mean ± SD	Mean ± SD	Mean ± SD	Mean ± SD
Number of points	271.0 ± 947.3	215.7 ± 819.0	133.2 ± 709.6	557.3 ± 1301.4
Mean VM (counts·s ⁻¹)	66.7 ± 44.9	50.9 ± 50.9	63.6 ± 44.7	86.0 ± 28.6

Abbreviation: VM, vector magnitude.

vector magnitude counts) and brick surfaces, concrete surfaces, greenness, condition, and cleanliness of equipment, covered areas, access points, and seated areas (regression coefficients are illustrated in Figure 4 and reported in Table S1). Significant positive associations were detected for sand, redness, blueness, colorful equipment, fields, and distance to road. However, when fitting one model including all exposures as independent

variables, the significant negative associations that remained were redness and cleanliness, while the significant positive associations were colorfulness and distance to road (Figure 5 and Table S2).

Regression modeling separately detected significant negative associations between time spent (number of GPS points) and

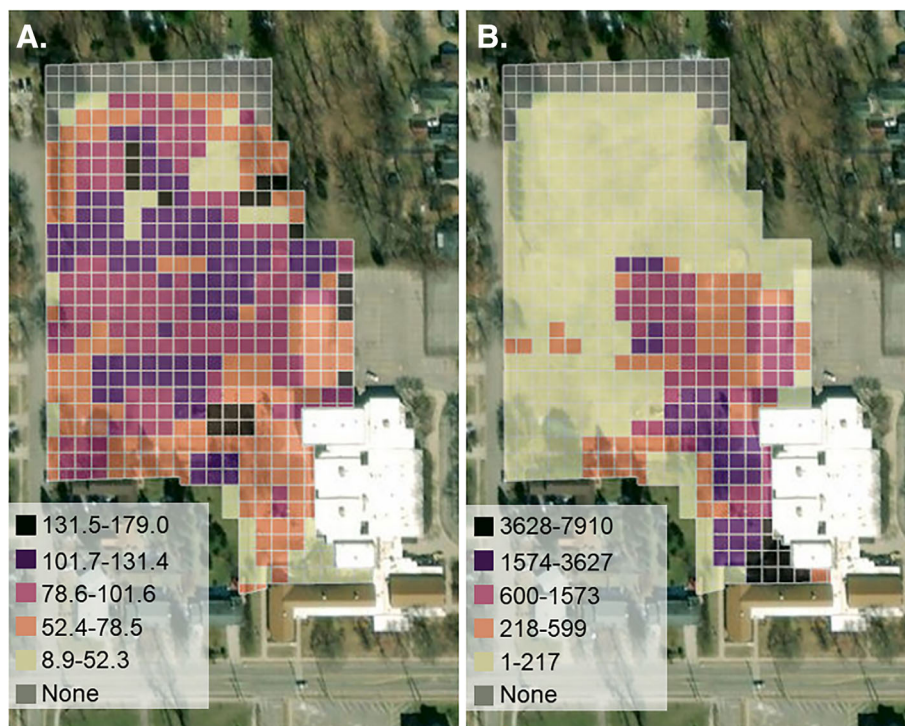


FIGURE 3 | For each audit grid cell at one school, mean vector magnitude counts-s⁻¹ (A) and total number of points (B).

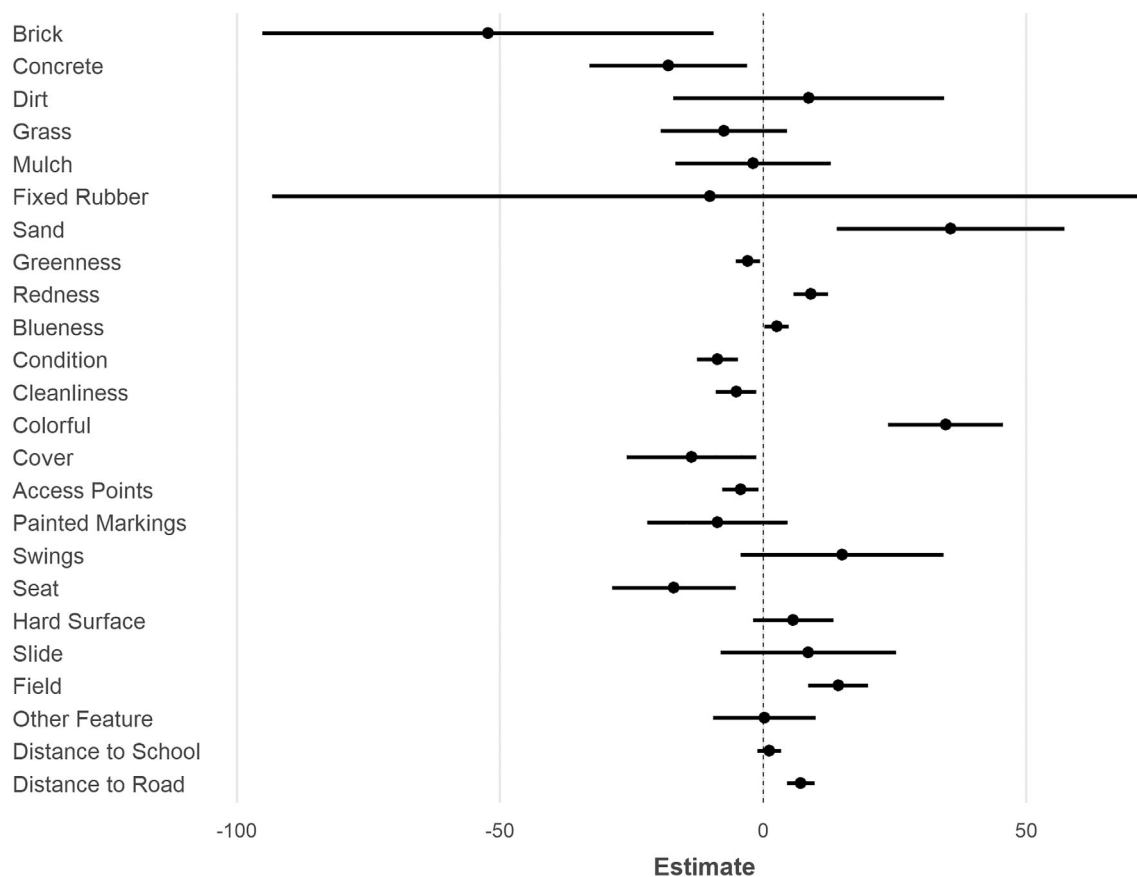


FIGURE 4 | Associations between exposures and mean vector magnitude counts-s⁻¹. Each exposure entered into a separate model. Continuous variables are mean-centered and scaled by one standard deviation.

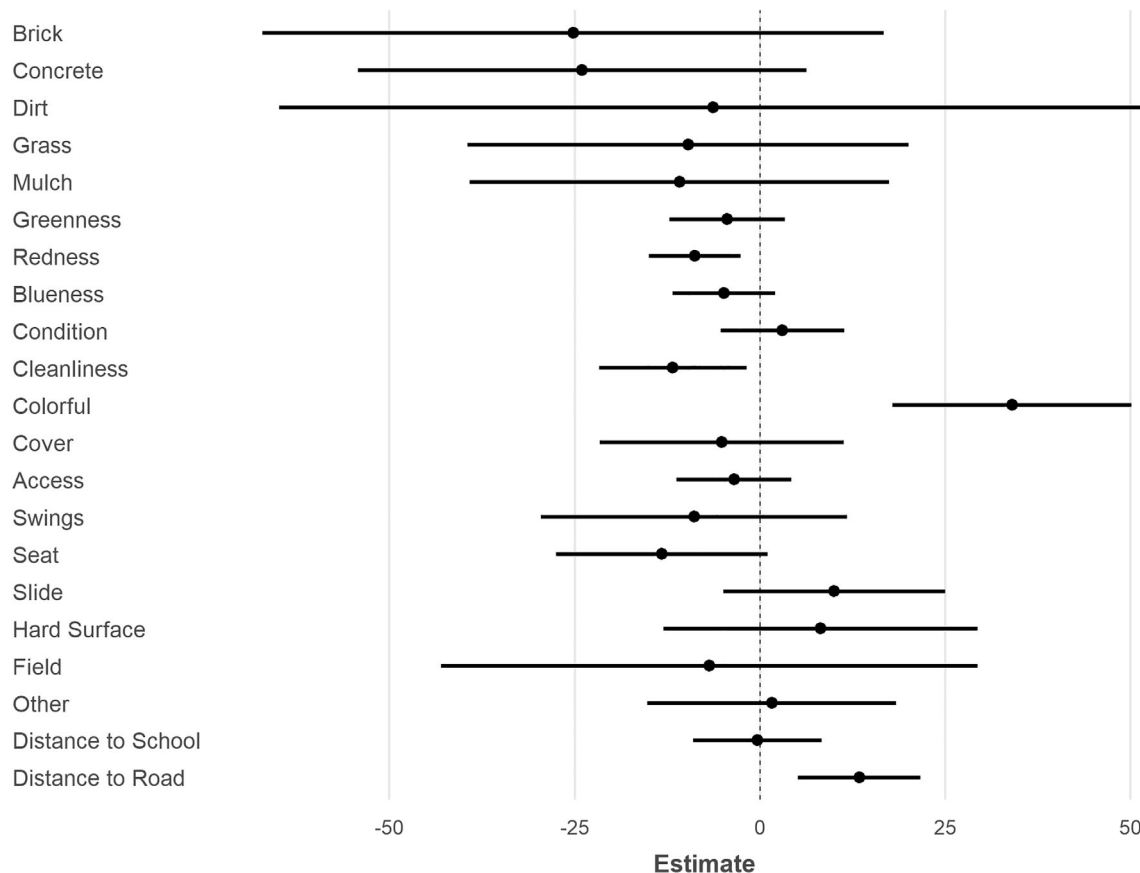


FIGURE 5 | Associations between exposures and mean vector magnitude counts·s⁻¹. All exposures included in a single model. Continuous variables are mean-centered and scaled by one standard deviation.

grass surfaces, sand, greenness, cleanliness of equipment, fields, and distance to school (regression coefficients are illustrated in Figure 6 and reported in Table S1). Significant positive associations were detected for mulch, redness, blueness, access points, swings, seats, hard surfaces, slide, other features, and distance to road. However, when fitting one model including all exposures as independent variables, the significant negative associations that remained were greenness and fields, while the significant positive association detected was condition of equipment (Figure 7 and Table S2).

4 | Discussion

We developed and applied a novel audit method designed to aid in understanding the inter-relationships between schoolyard features and child outcomes and health-related behaviors, which hopefully will be foundational in geospatial, disparities, and intervention-based research. SLIDE improves on existing methods for characterizing the schoolyard environment by simultaneously capturing multiple dimensions of schoolyard features and spatial information and allowing linkage with myriad data sources. In the present study, we linked SLIDE data to accelerometer and GPS device data regarding where children spend their time and are physically active. Our findings of associations between schoolyard features with these outcomes are mostly in line with prior research, indicating the audit method performed as expected. However, the SLIDE method could be

used to capture additional exposures that we did not include (e.g., noise, shade, microbial samples) and other outcomes (e.g., creative play captured via video and then spatially linked), so there are numerous potential future research directions worth considering to answer important research questions related to how schoolyards influence child social, cognitive, and physical development.

We found features positively and inversely related to the number of data points (i.e., time spent in a space). But, it is important to note that findings are drawn from a relatively small sample of 91 children across three schools. Features positively related to the number of data points were primarily significant in the individual models and mostly align with prior research. Specifically, swings [51], hard surface areas [39, 40, 47, 48, 51, 52], and mulch, slide, and other, all of which are linked with fixed equipment [50], were positively associated with the number of data points. There was also an inverse association of cleanliness with the number of data points, in line with Colabianchi, Maslow [78]. However, we found an inverse association of greenness and fields with the number of data points in both the individual and full models (also grass in the individual models) which is in contrast with prior research by Dymont, Bell [40]. Many of the green, grassy, or field areas in our study were far from the school or located in a forested belt around the schoolyard and may not have been accessible to children (see Figure S10). This is also in line with our finding of a negative association of distance to the school building with the number of data points (i.e., kids spent less time farther from the school)

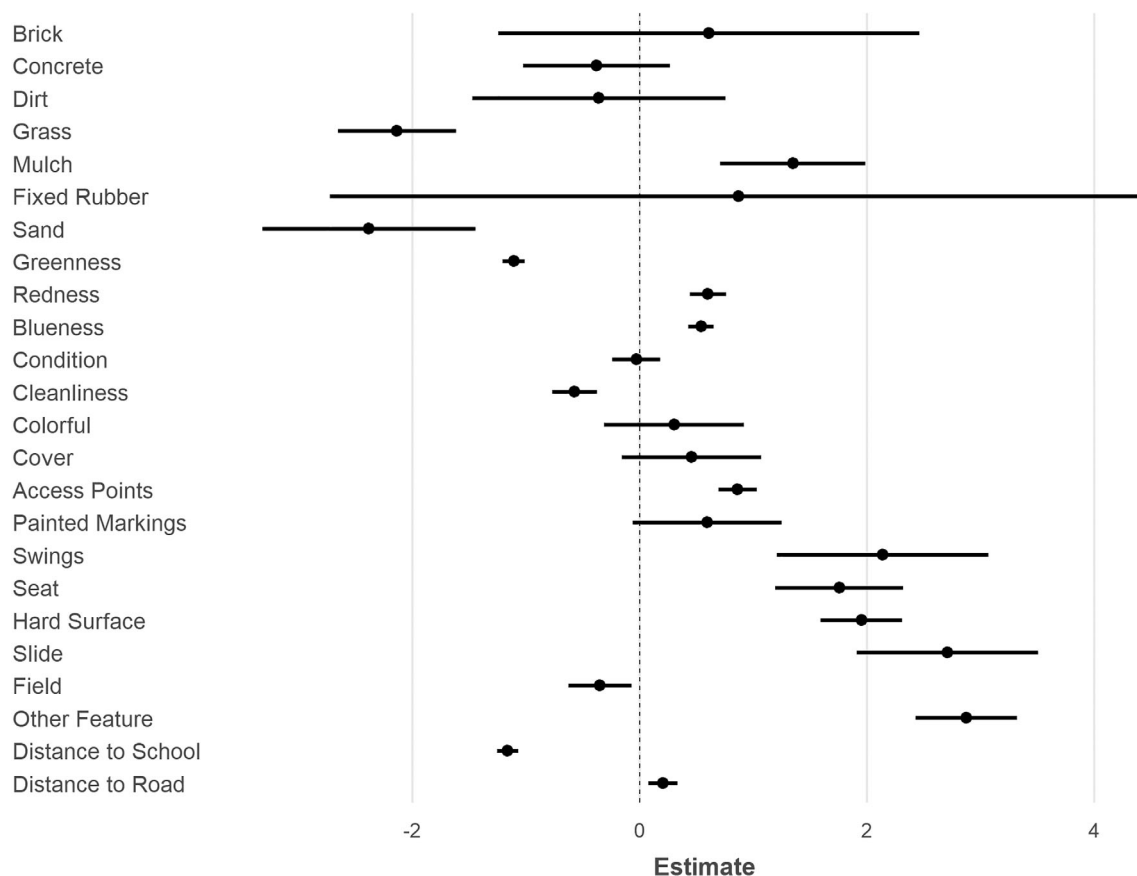


FIGURE 6 | Associations between exposures and number of points within each audit grid cell (log transformed)—as a measure of time spent. Each exposure entered into a separate model. Continuous variables are mean-centered and scaled by one standard deviation.

and with our finding of a positive association with redness, as the school buildings had red brick facades (see Figure S11). Further, blueness was also positively associated with time spent, and this likely relates to locations with more open views (not next to the school building or in an area with lots of trees, see Figure S12).

We also found factors positively and inversely associated with physical activity intensity (based on vector magnitude counts). In line with prior research, there was a positive association in the individual models of physical activity intensity with fields [49, 50, 79] and an inverse association with seating [41, 80, 81]. Seating was often placed on brick or concrete surfaces, which likely explains the inverse association of these features with physical activity, which was ameliorated in the fully controlled model. However, our findings of no association are in contrast to prior research that indicated fixed equipment [49, 51, 52] and blacktop or asphalt courts [39] also promote physical activity. This discrepancy may be related to the purpose of SLIDE versus other research. Specifically, SLIDE examined whether specific pieces of equipment (e.g., slide) were associated with physical activity in that particular spatial location. In contrast, other research has examined whether having a slide in any spatial location is related to physical activity (regardless of whether children are actually being active/playing in that location or not) or has aggregated all fixed equipment in analyses of child activity. Such studies highlight the complexity inherent in studying schoolyard design. Other associations identified in the present study, such as the

positive association of colorfulness and physical activity, have not yet been shown in prior research but could have implications for schoolyard design, if confirmed by further research.

While many of our findings align with prior research, we note that SLIDE also provides new insights. Chiefly, the prior research to which we compared our findings primarily associated the presence/absence of environmental features with children's behavior and is therefore most comparable to our separate regression models (with a single exposure or predictor). However, as demonstrated by our models including all exposures as independent variables, SLIDE facilitates a better understanding of the complex inter-relationships between schoolyard features and outcomes. For example, once we controlled for other characteristics like the presence of seating, certain ground covers where seating is often located were no longer significantly inversely related to physical activity intensity. On a larger sample of more diverse schoolyards, this would allow researchers to identify which specific features should be the focus of future schoolyard designs or renovations (e.g., seating versus ground cover).

The exposures captured in the present study were based on existing audit tools or prior research. However, researchers can use the SLIDE approach to collect information about other features of interest collected in person or using a device (e.g., noise, pollution, or shade) or via GIS (e.g., centrality or number of adjacent settings). A review of audit tools [43] illustrates several other items of potential interest, such as graffiti or litter.

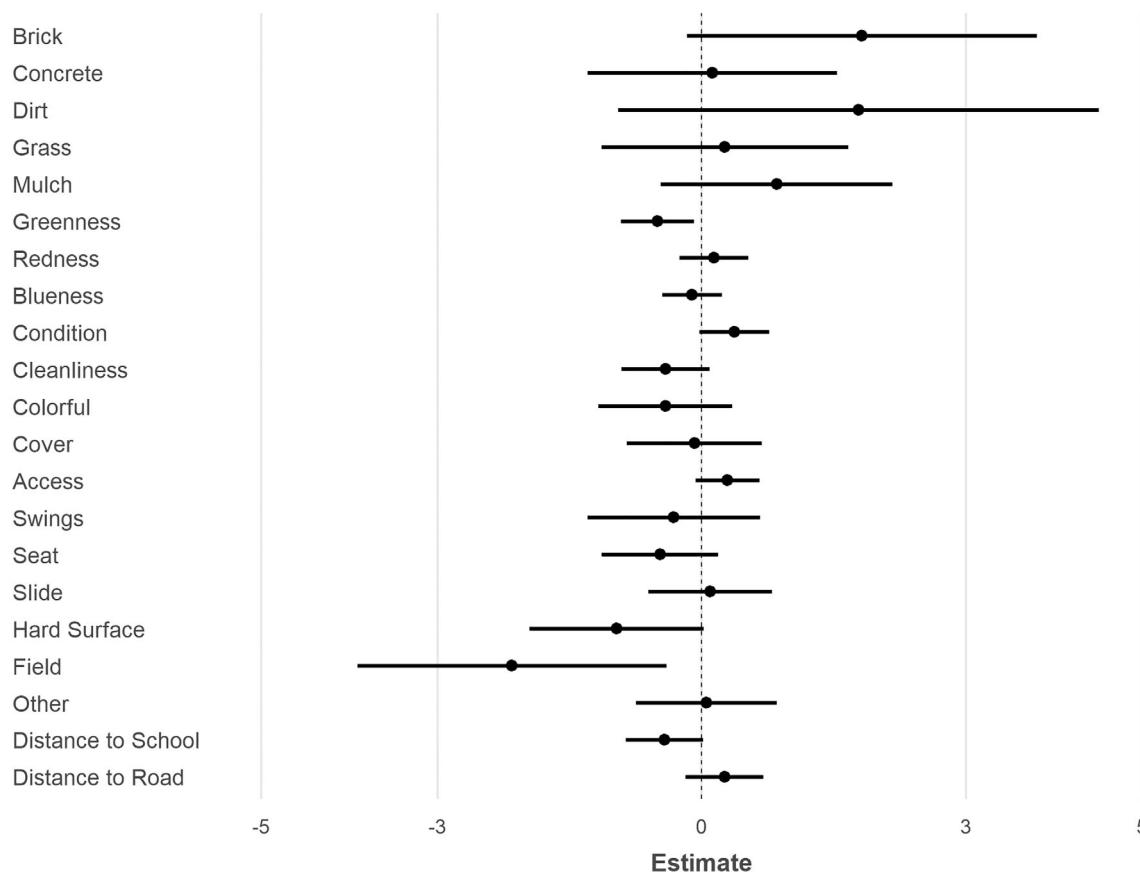


FIGURE 7 | Associations between exposures and number of points within each audit grid cell (log transformed). All exposure entered into a single model. Continuous variables are mean-centered and scaled by one standard deviation.

Because SLIDE data are spatially linked, one could use these data to study spatial variables (e.g., clusters of play-promoting features, centrality) [73]. SLIDE audit data can be collated with other data at the child level like age or sex, the school level like schoolyard size or number of children, or temporal characteristics like time of year/weather. Outcomes other than physical activity or time could be explored, like type of play (e.g., risky play) or social interaction. An important future application of SLIDE is to better understand and address inequities [35, 36] so we note that SLIDE could be used to quantify this inequity and interventions aiming to reduce it, such as greening or renovations. Additional exposures like views of blight or neighborhood disorder from the schoolyard could also be relevant for this purpose.

As this is the first study employing SLIDE, it is relevant to overview a few ‘lessons learned.’ First, some schoolyards or areas of the schoolyard may not require such fine resolution (7m) and this would shorten data collection time and data processing time. The resolution of SLIDE must align with the scale of the study area to ensure that it captures relevant data without introducing unnecessary generalization. For instance, a resolution that is too coarse for a small, dynamic area (e.g., one with significant variations in slope or vegetation), would lead to a loss of important detail. In areas where there are fewer dynamic variables (e.g., a flat, homogeneous field), a coarser resolution can still provide sufficient information without compromising the accuracy of the analysis. Second, some 360° cameras now

include built-in GPS devices, which could save data collection time and facilitate the processing and spatial linkage of data from multiple sources. Third, we recommend visiting the schoolyard and delimiting the boundaries using a GPS device before carrying out the audit method. This will allow the researchers to estimate the amount of time required and to generate the data collection grid precisely prior to data collection. In the present study, two of the three schoolyards were located next to public parks and it was unclear whether children were allowed to access these spaces. This resulted in unnecessary time spent auditing.

For those interested in employing this audit method, a key first step for SLIDE users is to decide on audit measures of interest. Existing coding schemes should be used when possible (e.g., condition and cleanliness in the present study were based on EAPRS). If new coding schemes must be developed, definitions and rules should be thoroughly reported. SLIDE users should report: metadata for all existing data layers, coding schemes for all user-coded variables, how long the process took (minutes/m²), what season and time of day the method was conducted, how many researchers were used (as an indicator of reliability), which features were recorded, and the make and model of all devices used.

Our findings also point to some practical implications for schoolyard design. Our findings indicate that colorful (2 or more non-neutral colors) and well-maintained equipment, located

farther from roads, may be important dimensions of schoolyard design. Portable equipment (e.g., balls) or school rules (e.g., where children are allowed to play) may influence the use of open, green field spaces but were not captured in the present study.

This study and the audit method are not without limitations. While our purpose was to illustrate the use of SLIDE and not to thoroughly investigate the interrelationships amongst environmental features, we still collected cross-sectional data at a small number of similar schools with a majority White student population, which can all be considered limitations. Further research is needed in lower socioeconomic areas, more dense urban areas, on a variety of schoolyards with varying sizes and distances to roads, and variation in school recess policies, while also considering seasonal changes and sample characteristics. Because of the relatively small study of 91 children across three schools showed little variation in several of the outcomes, findings must be interpreted with caution. Future research applying this method in intervention studies is also needed. For example, SLIDE may be useful when evaluating the effectiveness of and the specific areas/features of the schoolyard that were associated with changes in recess physical activity following schoolyard interventions [13], notable schoolyard greening initiatives [82]. Our use of a hip-worn accelerometer is a limitation as it may not capture some schoolyard activities such as swinging, monkey bars, or activities that only use upper-body movements (e.g., tetherball). Finally, the audit method is time intensive and requires technical skills (e.g., 360 camera, GPS device); however, it provides rich information regarding the schoolyard environment and can be modified as necessary. While we used software that costs money, SLIDE could be employed using Google Earth or Google Street View, QField, or paper forms to reduce time and financial burden to researchers [58]. However, the imagery in Google Earth and Google Street View poses other challenges including recency, distance from schoolyard, and image resolution for smaller schoolyard features [58].

5 | Conclusion

The SLIDE audit method can usefully be employed to understand the inter-relationships between schoolyard features and child outcomes and health-related behaviors. The method provides a framework for linking data from in-person (or virtual) audits, device-based measures, and GIS. It requires a small number of data collectors (depending on the scope of the features collected) and a time commitment similar to or less than that of other auditing tools [84]. This method may be foundational to answering numerous research questions in geospatial, disparities, and intervention-based research for child health promotion and equity.

Conflicts of Interest

The authors declare no conflicts of interest.

References

1. B. Hyndman, *Contemporary School Playground Strategies for Healthy Students* (Springer, 2017).

2. S. Herrington and M. Brussoni, "Beyond Physical Activity: The Importance of Play and Nature-Based Play Spaces for Children's Health and Development," *Current Obesity Reports* 4, no. 4 (2015): 477–483.
3. T. Skelton and S. C. Aitken, *Establishing Geographies of Children and Young People* (Springer, 2019).
4. P. Gil-Madrona, M. Martínez-López, A. Prieto-Ayuso, et al., "Contribution of Public Playgrounds to Motor, Social, and Creative Development and Obesity Reduction in Children," *Sustainability* 11, no. 14 (2019): 3787.
5. R. R. Suminski, D. Ding, R. Lee, L. May, T. Tota, and D. Dinius, "Youth Physical Activity Opportunities in Lower and Higher Income Neighborhoods," *Journal of Urban Health* 88, no. 4 (2011): 599–615.
6. M. M. Scott, D. A. Cohen, K. R. Evenson, et al., "Weekend Schoolyard Accessibility, Physical Activity, and Obesity: The Trial of Activity in Adolescent Girls (TAAG) Study," *Preventive Medicine* 44, no. 5 (2007): 398–403.
7. O. Jarrett, "A Research-Based Case for Recess: Position Paper," US Play Coalition with American Association for the Child's Right to Play. (2019).
8. C. Ramstetter and R. Murray, "Time to Play: Recognizing the Benefits of Recess," *American Educator* 41, no. 1 (2017): 17.
9. C. M. Bohn-Gettler and A. D. Pellegrini, "Recess in Primary School: The Disjuncture Between Educational Policy and Scientific Research," in *Justice, Conflict and Wellbeing* (Springer, 2014), 313–336.
10. B. C. Guinhouya, M. Lemdani, G. K. Apété, A. Durocher, C. Vilhelm, and H. Hubert, "How School Time Physical Activity Is the 'Big One' for Daily Activity Among Schoolchildren: A Semi-Experimental Approach," *Journal of Physical Activity and Health* 6, no. 4 (2009): 510–519.
11. K. Broekhuizen, A.-M. Scholten, and S. I. De Vries, "The Value of (Pre) School Playgrounds for Children's Physical Activity Level: A Systematic Review," *International Journal of Behavioral Nutrition and Physical Activity* 11, no. 1 (2014): 1–28.
12. K. A. Clevenger, M. J. Wierenga, C. A. Howe, and K. A. Pfeiffer, "A Systematic Review of Child and Adolescent Physical Activity by Schoolyard Location," *Kinesiology Review* 9, no. 2 (2020): 147–158, <https://doi.org/10.1123/kr.2019-0009>.
13. J. C. Bikomeye, J. Balza, and K. M. Beyer, "The Impact of Schoolyard Greening on Children's Physical Activity and Socioemotional Health: A Systematic Review of Experimental Studies," *International Journal of Environmental Research and Public Health* 18, no. 2 (2021): 535.
14. C. R. Bates, A. M. Bohnert, and D. E. Gerstein, "Green Schoolyards in Low-Income Urban Neighborhoods: Natural Spaces for Positive Youth Development Outcomes," *Frontiers in Psychology* 9 (2018): 805.
15. W. V. Massey, M. B. Stellino, and M. Fraser, "Individual and Environmental Correlates of School-Based Recess Engagement," *Preventive Medicine Reports* 11 (2018): 247–253.
16. W. V. Massey, M. B. Stellino, S. P. Mullen, J. Claassen, and M. Wilkison, "Development of the Great Recess Framework—Observational Tool to Measure Contextual and Behavioral Components of Elementary School Recess," *BMC Public Health* 18, no. 1 (2018): 1–11.
17. J. Juhola, C. G. Magnussen, J. S. Viikari, et al., "Tracking of Serum Lipid Levels, Blood Pressure, and Body Mass Index From Childhood to Adulthood: The Cardiovascular Risk in Young Finns Study," *Journal of Pediatrics* 159, no. 4 (2011): 584–590.
18. R. Telama, "Tracking of Physical Activity From Childhood to Adulthood: A Review," *Obesity Facts* 2, no. 3 (2009): 187–195.
19. W. V. Massey, A. Szarabajko, J. Thalken, D. Perez, and S. P. Mullen, "Memories of School Recess Predict Physical Activity Enjoyment and Social-Emotional Well-Being in Adults," *Psychology of Sport and Exercise* 55 (2021): 101948, <https://doi.org/10.1016/j.psychsport.2021.101948>.

20. V. Nicaise, D. Kahan, K. Reuben, and J. F. Sallis, "Evaluation of a Redesigned Outdoor Space on Preschool Children's Physical Activity During Recess," *Pediatric Exercise Science* 24, no. 4 (2012): 507–518.
21. P. Anthamatten, L. Brink, S. Lampe, E. Greenwood, B. Kingston, and C. Nigg, "An Assessment of Schoolyard Renovation Strategies to Encourage Children's Physical Activity," *International Journal of Behavioral Nutrition and Physical Activity* 8, no. 1 (2011): 1–9, <https://doi.org/10.1186/1479-5868-8-27>.
22. J. E. van Dijk-Wesselius, J. Maas, M. van Vugt, and A. E. van den Berg, "A Comparison of Children's Play and Non-Play Behavior Before and After Schoolyard Greening Monitored by Video Observations," *Journal of Environmental Psychology* 80 (2022): 101760.
23. M. A. Raney, C. F. Hendry, and S. A. Yee, "Physical Activity and Social Behaviors of Urban Children in Green Playgrounds," *American Journal of Preventive Medicine* 56, no. 4 (2019): 522–529.
24. D. R. Bassett, E. C. Fitzhugh, G. W. Heath, et al., "Estimated Energy Expenditures for School-Based Policies and Active Living," *American Journal of Preventive Medicine* 44, no. 2 (2013): 108–113.
25. A. Zhang, R. Bokel, A. van den Dobbelsteen, Y. Sun, Q. Huang, and Q. Zhang, "An Integrated School and Schoolyard Design Method for Summer Thermal Comfort and Energy Efficiency in Northern China," *Building and Environment* 124 (2017): 369–387.
26. S. G. Danks, *Asphalt to Ecosystems: Design Ideas for Schoolyard Transformation* (New Village Press, 2010).
27. A. Kopeva, O. Khrapko, and O. Ivanova, "Landscape Planning of Schoolyards," in *IOP Conference Series: Materials Science and Engineering* (IOP Publishing, 2017).
28. P. Dadvand, I. Rivas, X. Basagaña, et al., "The Association Between Greenness and Traffic-Related Air Pollution at Schools," *Science of the Total Environment* 523 (2015): 59–63.
29. M. C. Minguillón, I. Rivas, T. Moreno, et al., "Road Traffic and Sandy Playground Influence on Ambient Pollutants in Schools," *Atmospheric Environment* 111 (2015): 94–102.
30. J. K. Vanos, "Children's Health and Vulnerability in Outdoor Microclimates: A Comprehensive Review," *Environment International* 76 (2015): 1–15.
31. C. Kelz, G. W. Evans, and K. Röderer, "The Restorative Effects of Redesigning the Schoolyard: A Multi-Methodological, Quasi-Experimental Study in Rural Austrian Middle Schools," *Environment and Behavior* 47, no. 2 (2015): 119–139.
32. K. T. Stevenson, R. Moore, N. Cosco, et al., "A National Research Agenda Supporting Green Schoolyard Development and Equitable Access to Nature," *Elementa: Science of the Anthropocene* 8, no. 1 (2020): 1–11.
33. F. Baró, D. A. C. Caballero, C. P. del Pulgar, M. Triguero-Mas, and I. Anguelovski, "School Greening: Right or Privilege? Examining Urban Nature Within and Around Primary Schools Through an Equity Lens," *Landscape and Urban Planning* 208 (2021): 104019.
34. G. W. Evans and E. Kantrowitz, "Socioeconomic Status and Health: The Potential Role of Environmental Risk Exposure," *Annual Review of Public Health* 23, no. 1 (2002): 303–331.
35. C. Arroyo-Johnson, K. Woodward, L. Milam, et al., "Still Separate, Still Unequal: Social Determinants of Playground Safety and Proximity Disparities in St. Louis," *Journal of Urban Health* 93, no. 4 (2016): 627–638, <https://doi.org/10.1007/s11524-016-0063-8>.
36. C. Buck, A. Bolbos, and S. Schneider, "Do Poorer Children Have Poorer Playgrounds? A Geographically Weighted Analysis of Attractiveness, Cleanliness, and Safety of Playgrounds in Affluent and Deprived Urban Neighborhoods," *Journal of Physical Activity and Health* 16, no. 6 (2019): 397–405.
37. P. F. Saint-Maurice, G. J. Welk, P. Silva, M. Siahpush, and J. Huberty, "Assessing Children's Physical Activity Behaviors at Recess: A Multi-Method Approach," *Pediatric Exercise Science* 23, no. 4 (2011): 585–599.
38. C. S. Pawlowski, H. B. Andersen, J. Troelsen, and J. Schipperijn, "Children's Physical Activity Behavior During School Recess: A Pilot Study Using GPS, Accelerometer, Participant Observation, and Go-Along Interview," *PLoS One* 11, no. 2 (2016): e0148786.
39. I. E. Black, N. N. Menzel, and T. J. Bungum, "The Relationship Among Playground Areas and Physical Activity Levels in Children," *Journal of Pediatric Health Care* 29, no. 2 (2015): 156–168.
40. J. E. Dymont, A. C. Bell, and A. J. Lucas, "The Relationship Between School Ground Design and Intensity of Physical Activity," *Children's Geographies* 7, no. 3 (2009): 261–276.
41. T. A. Farley, R. A. Meriwether, E. T. Baker, J. C. Rice, and L. S. Webber, "Where Do the Children Play? The Influence of Playground Equipment on Physical Activity of Children in Free Play," *Journal of Physical Activity & Health* 5, no. 2 (2008): 319–331.
42. L. J. Willenberg, R. Ashbolt, D. Holland, et al., "Increasing School Playground Physical Activity: A Mixed Methods Study Combining Environmental Measures and Children's Perspectives," *Journal of Science and Medicine in Sport* 13, no. 2 (2010): 210–216.
43. H. G. Lane, H. G. Calvert, R. Deitch, et al., "A Systematic Review of Existing Observational Tools to Measure the Food and Physical Activity Environment in Schools," *Health & Place* 66 (2020): 102388.
44. I. Fjørtoft, B. Kristoffersen, and J. Sageie, "Children in Schoolyards: Tracking Movement Patterns and Physical Activity in Schoolyards Using Global Positioning System and Heart Rate Monitoring," *Landscape and Urban Planning* 93, no. 3–4 (2009): 210–217.
45. I. Fjørtoft, O. Löfman, and K. Halvorsen Thorén, "Schoolyard Physical Activity in 14-Year-Old Adolescents Assessed by Mobile GPS and Heart Rate Monitoring Analysed by GIS," *Scandinavian Journal of Public Health* 38, no. 5_suppl (2010): 28–37.
46. H. B. Andersen, L. B. Christiansen, C. S. Pawlowski, and J. Schipperijn, "What We Build Makes a Difference—Mapping Activating Schoolyard Features After Renewal Using GIS, GPS and Accelerometers," *Landscape and Urban Planning* 191 (2019): 103617, <https://doi.org/10.1016/j.landurbplan.2019.103617>.
47. H. B. Andersen, C. D. Klinker, M. Toftager, C. S. Pawlowski, and J. Schipperijn, "Objectively Measured Differences in Physical Activity in Five Types of Schoolyard Area," *Landscape and Urban Planning* 134 (2015): 83–92.
48. K. A. Clevenger, G. Sinha, and C. A. Howe, "Comparison of Methods for Analyzing Global Positioning System and Accelerometer Data During School Recess," *Measurement in Physical Education and Exercise Science* 23, no. 1 (2019): 58–68.
49. L. A. Brink, C. R. Nigg, S. M. Lampe, B. A. Kingston, A. L. Mootz, and W. van Vliet, "Influence of Schoolyard Renovations on Children's Physical Activity: The Learning Landscapes Program," *American Journal of Public Health* 100, no. 9 (2010): 1672–1678.
50. A. E. Springer, Y. Tanguturi, N. Ranjit, K. A. Skala, and S. H. Kelder, "Physical Activity During Recess in Low-Income Third-Grade Texas Students," *American Journal of Health Behavior* 37, no. 3 (2013): 318–324.
51. P. Anthamatten, L. Brink, B. Kingston, E. Kutchman, S. Lampe, and C. Nigg, "An Assessment of Schoolyard Features and Behavior Patterns in Children's Utilization and Physical Activity," *Journal of Physical Activity and Health* 11, no. 3 (2014): 564–573.
52. C. A. Howe, K. A. Clevenger, B. Plow, S. Porter, and G. Sinha, "Using Video Direct Observation to Assess Children's Physical Activity During Recess," *Pediatric Exercise Science* 30, no. 4 (2018): 516–523.

53. D. S. Ward, S. Mazzucca, C. McWilliams, and D. Hales, "Use of the Environment and Policy Evaluation and Observation as a Self-Report Instrument (EPAO-SR) to Measure Nutrition and Physical Activity Environments in Child Care Settings: Validity and Reliability Evidence," *International Journal of Behavioral Nutrition and Physical Activity* 12, no. 1 (2015): 1.
54. B. E. Saelens, L. D. Frank, C. Auffrey, R. C. Whitaker, H. L. Burdette, and N. Colabianchi, "Measuring Physical Environments of Parks and Playgrounds: EAPRS Instrument Development and Inter-Rater Reliability," *Journal of Physical Activity and Health* 3, no. s1 (2006): S190–S207.
55. A. Kaczynski and S. W. Stanis, "Community Park Audit Tool (CPAT)," (2010).
56. R. E. Lee, K. M. Booth, J. Y. Reese-Smith, G. Regan, and H. H. Howard, "The Physical Activity Resource Assessment (PARA) Instrument: Evaluating Features, Amenities and Incivilities of Physical Activity Resources in Urban Neighborhoods," *International Journal of Behavioral Nutrition and Physical Activity* 2, no. 1 (2005): 1–9.
57. M. Perlman, G. L. Zellman, and V.-N. Le, "Examining the Psychometric Properties of the Early Childhood Environment Rating Scale-Revised (ECERS-R)," *Early Childhood Research Quarterly* 19, no. 3 (2004): 398–412.
58. A. Rzotkiewicz, A. L. Pearson, B. V. Dougherty, A. Shortridge, and N. Wilson, "Systematic Review of the Use of Google Street View in Health Research: Major Themes, Strengths, Weaknesses and Possibilities for Future Research," *Health & Place* 52 (2018): 240–246.
59. M. D. Bader, S. J. Mooney, Y. J. Lee, et al., "Development and Deployment of the Computer Assisted Neighborhood Visual Assessment System (CANVAS) to Measure Health-Related Neighborhood Conditions," *Health & Place* 31 (2015): 163–172, <https://doi.org/10.1016/j.healthplace.2014.10.012>.
60. S. J. Mooney, M. D. M. Bader, G. S. Lovasi, K. M. Neckerman, J. O. Teitler, and A. G. Rundle, "Validity of an Ecometric Neighborhood Physical Disorder Measure Constructed by Virtual Street Audit," *American Journal of Epidemiology* 180, no. 6 (2014): 626–635.
61. S. J. Lafontaine, M. Sawada, and E. Kristjansson, "A Direct Observation Method for Auditing Large Urban Centers Using Stratified Sampling, Mobile GIS Technology and Virtual Environments," *International Journal of Health Geographics* 16, no. 1 (2017): 1–15.
62. N. Wilson, G. Thomson, and R. Edwards, "The Potential of Google Street View for Studying Smokefree Signage," *Australian and New Zealand Journal of Public Health* 39, no. 3 (2015): 295–296.
63. D. K. Brown, J. L. Barton, and V. F. Gladwell, "Viewing Nature Scenes Positively Affects Recovery of Autonomic Function Following Acute-Mental Stress," *Environmental Science & Technology* 47 (2013): 5562–5569.
64. V. F. Gladwell, D. K. Brown, J. L. Barton, et al., "The Effects of Views of Nature on Autonomic Control," *European Journal of Applied Physiology* 112 (2012): 3379–3386.
65. H. P. Kotabe, O. Kardan, and M. G. Berman, "The Nature-Disorder Paradox: A Perceptual Study on How Nature Is Disorderly Yet Aesthetically Preferred," *Journal of Experimental Psychology: General* 146, no. 8 (2017): 1126–1142.
66. M. G. Berman, M. C. Hout, O. Kardan, et al., "The Perception of Naturalness Correlates With Low-Level Visual Features of Environmental Scenes," *PLoS One* 9, no. 12 (2014): e114572.
67. M. G. Berman, J. Jonides, and S. A. Kaplan, "The Cognitive Benefits of Interacting With Nature," *Psychological Science* 19, no. 12 (2008): 1207–1212.
68. S. A. Kaplan, "The Restorative Benefits of Nature: Toward an Integrative Framework," *Journal of Environmental Psychology* 15, no. 3 (1995): 169–182.
69. R. Ghayouri and S. Ayat, "Color Preference Based on Children's Choices for the Interior Spaces of Kindergarten," *International Research Journal of Advanced Engineering and Science* 5, no. 2 (2020): 154–159.
70. C. J. Boyatzis and R. Varghese, "Children's Emotional Associations With Colors," *Journal of Genetic Psychology* 155, no. 1 (1994): 77–85.
71. A. V. Fisher, K. E. Godwin, and H. Seltman, "Visual Environment, Attention Allocation, and Learning in Young Children: When Too Much of a Good Thing May be Bad," *Psychological Science* 25, no. 7 (2014): 1362–1370.
72. W. R. Smith, R. Moore, N. Cosco, et al., "Increasing Physical Activity in Childcare Outdoor Learning Environments: The Effect of Setting Adjacency Relative to Other Built Environment and Social Factors," *Environment and Behavior* 48, no. 4 (2016): 550–578.
73. K. A. Clevenger, K. A. Pfeiffer, and A. H. K. Montoye, "Cross-Generational Comparability of Hip- and Wrist-Worn ActiGraph GT3X+, wGT3X-BT, and GT9X Accelerometers During Free-Living in Adults," *Journal of Sports Sciences* 38, no. 24 (2020): 2794–2802.
74. K. A. Clevenger, K. A. Pfeiffer, and A. H. K. Montoye, "Cross-Generational Comparability of Raw and Count-Based Metrics From ActiGraph GT9X and wGT3X-BT Accelerometers During Free-Living in Youth," *Measurement in Physical Education and Exercise Science* 24, no. 3 (2020): 194–204.
75. J. H. Migueles, C. Cadenas-Sanchez, U. Ekelund, et al., "Accelerometer Data Collection and Processing Criteria to Assess Physical Activity and Other Outcomes: A Systematic Review and Practical Considerations," *Sports Medicine* 47, no. 9 (2017): 1821–1845.
76. J. J. McClain, T. L. Abraham, T. A. Brusseau, and C. Tudor-Locke, "Epoch Length and Accelerometer Outputs in Children: Comparison to Direct Observation," *Medicine and Science in Sports and Exercise* 40, no. 12 (2008): 2080–2087, <https://doi.org/10.1249/MSS.0b013e3181824d98>.
77. N. Colabianchi, A. L. Maslow, and K. Swayampakala, "Features and Amenities of School Playgrounds: A Direct Observation Study of Utilization and Physical Activity Levels Outside of School Time," *International Journal of Behavioral Nutrition and Physical Activity* 8, no. 1 (2011): 1–11.
78. C. Wood, V. Gladwell, and J. Barton, "A Repeated Measures Experiment of School Playing Environment to Increase Physical Activity and Enhance Self-Esteem in UK School Children," *PLoS One* 9, no. 9 (2014): e108701.
79. W. R. Smith, R. Moore, N. Cosco, et al., "Increasing Physical Activity in Childcare Outdoor Learning Environments: The Effect of Setting Adjacency Relative to Other Built Environment and Social Factors," *Environment and Behavior* 48, no. 4 (2016): 550–578.
80. N. G. Cosco, R. C. Moore, and M. Z. Islam, "Behavior Mapping: A Method for Linking Preschool Physical Activity and Outdoor Design," *Medicine & Science in Sports & Exercise* 42, no. 3 (2010): 513–519.
81. A. M. Parrish, K. H. Chong, A. L. Moriarty, M. Batterham, and N. D. Ridgers, "Interventions to Change School Recess Activity Levels in Children and Adolescents: A Systematic Review and Meta-Analysis," *Sports Medicine* 50, no. 12 (2020): 2145–2173.
82. C. D. Furr-Holden, K. D. Campbell, A. J. Milam, M. J. Smart, N. A. Ialongo, and P. J. Leaf, "Metric Properties of the Neighborhood Inventory for Environmental Typology (NIETy): An Environmental Assessment Tool for Measuring Indicators of Violence, Alcohol, Tobacco, and Other Drug Exposures," *Evaluation Review* 34, no. 3 (2010): 159–184, <https://doi.org/10.1177/0193841X10368493>.

Supporting Information

Additional supporting information can be found online in the Supporting Information section.